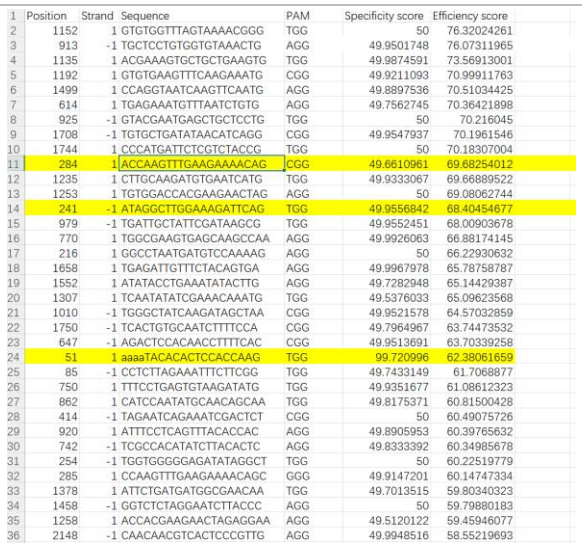
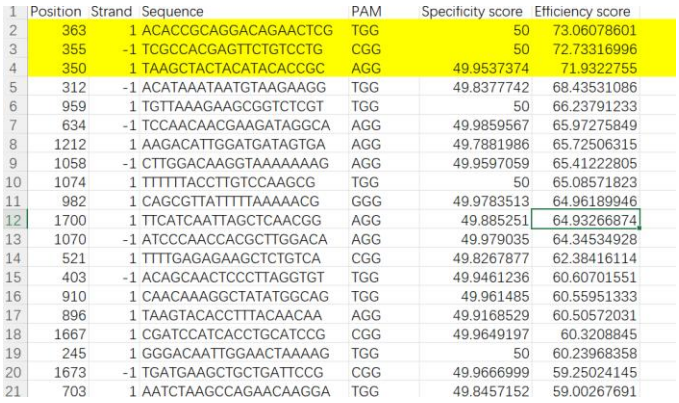
Notebook on yeast gene knockout experiments

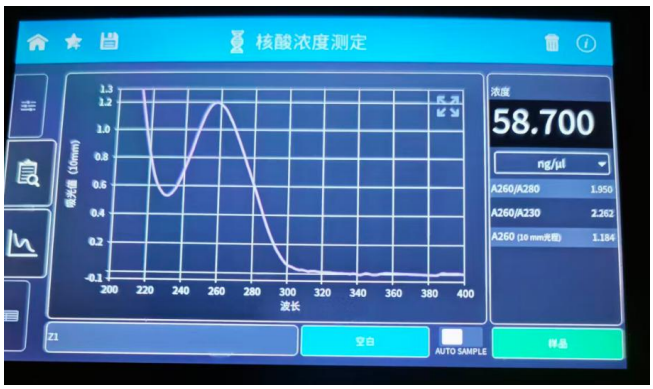
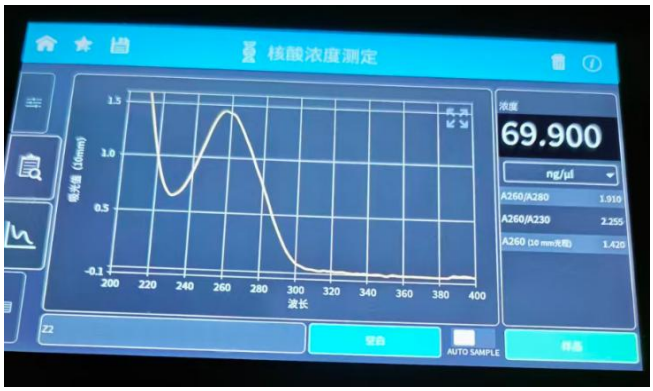
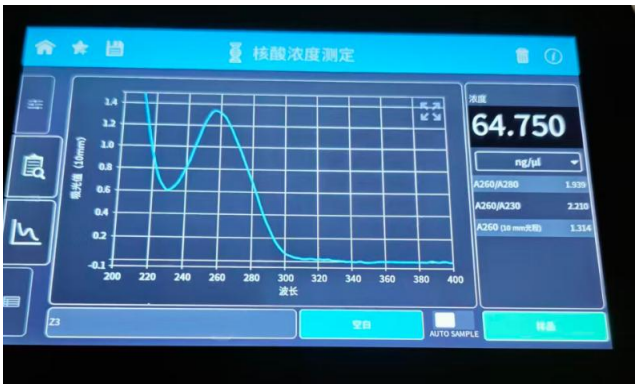
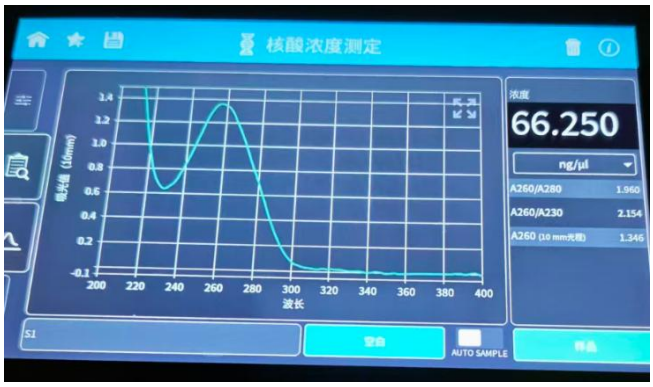
No 1 Date：April 1 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

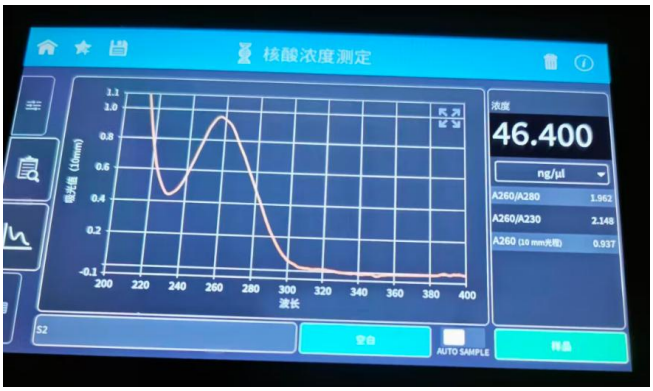
Experiment	SgRNA design for gene knockdown in yeast
Note	SGD（https://www.yeastgenome.org/） Benchling（https://www.benchling.com）
Result	>GPA1 YHR005C SGDID:S000001047 MGCTVSTQTIGDESDPFLQNKRANDVIEQSLQLEKQRDKNEIKLLLLGAG ESGKSTVLKQLKLLHQGGFSHQERLQYAQVIWADAIQSMKILIIQARKLGI QLDCDDPINNKDLFACKRILLKAKALDYINASVAGGSDFLNDYVLKYSER YETRRRVQSTGRAKAAFDEDGNISNVKSDTDRDAETVTQNEDADRNNSS RINLQDICKDLNQEGDDQMFVRKTSREIQGQNRRNLIHEDIAKAIKQLWN NDKGIKQCFARSNEFQLEGSAAYYFDNIEKFASPNYVCTDEDILKGRIKTT GITETEFNIGSSKFKVLDAGGQRSERKKWIHCFEGITAVLFVLAMSEYDQM LFEDERVNRMHESIMLFDTLLNSKWFKDTPFILFLNKIDLFEEKVKSMPIR KYFPDYQGRVGDAEAGLKYFEKIFLSLNKTNKPIYVKRTCATDTQTMKFV LSAVTDLIIQQNLKKIGII* **Gpa1 protein sequence** >SST2 YLR452C SGDID:S000004444 MVDKNRTLHELSSKNFSRTPNGLIFTNDLKTVYSIFLICLDLKEKKHSSDT KSFLLTAFTKHFHFTFTYQEAIKAMGQLELKVDMNTTCINVSYNIKPSLAR HLLTLFMSSKLLHTPQDRTRGEPKEKVLFQPTPKGVAVLQKYVRDIGLKT MPDILLSSFNSMKLFTFERSSVTDSIIHSDYLIHILFIKMMGAKPNVWSPTN ADDPLPCLSSLLEYTNNDDTFTFEKSKPEQGWQAQIGNIDINDLERVSPLA HRFFTNPDSESHTQYYVSNAGIRLFENKTFGTSKKIVIKYTFTTKAIWQWI MDCTDIMHVKEAVSLAALFLKTGLIVPVLLQPSRTDKKKFQISRSSFFTLS KRGWDLVSWTGCKSNNIRAPNGSTIDLDFTLRGHMTVRDEKKTLDDSEG FSQDMLISSSNLNKLDYVLTDPGMRYLFRRHLEKELCVENLDVFIEIKRFL KKMTILKKLIDSKHCDKKSNTSTSKNNIVKTIDSALMKQANECLEMAYHI YSSYIMIGSPYQLNIHHNLRQNISDIMLHPHSPLSEHFPTNLYDPSPASAESA ASSISSTEADTLGEPPEVSLKPSKNLSNENCSFKKQGFKHQLKEYKPAPLTL AETHSPNASVENSHTIVRYGMDNTQNDTKSVESFPATLKVLRKLYPLFEIV SNEMYRLMNNDSFQKFTQSDVYKDASALIEIQEKC* **Sst2 protein sequence**

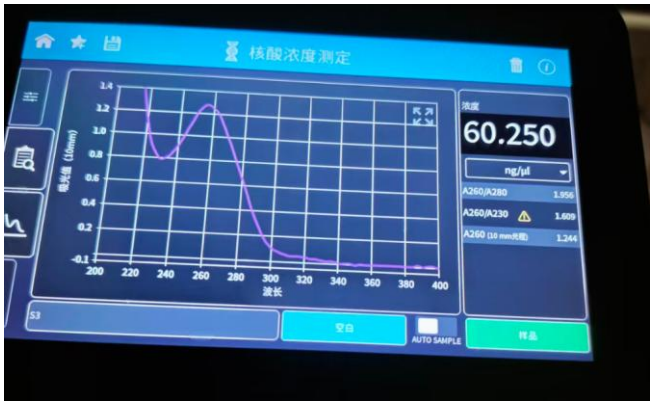
	 **SgRNA of Far1**  **SgRNA of sst2**
Explanation	In this experiment, the genomic DNA sequences, CDs and corresponding amino acid sequences of sst2 and far1 genes were successfully obtained through SGD database, ensuring the accuracy and authority of sequence sources. At the same time, sgRNAs with high specificity and efficiency scores were designed for the above target genes by using the benchling platform, which laid the foundation for subsequent CRISPR experiments.
Next plan	synthesize oligonucleotides and start wet-lab experiments.

No 2 Date：May 5–9, 2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

Experiment	Recovery of Glycerol Stocks and Plasmid Extraction of pML104
Note	Protocol 3: Plasmid Extraction from E. coli

Result	   





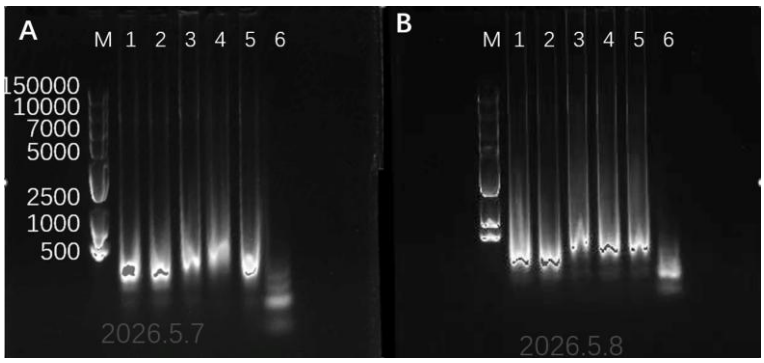
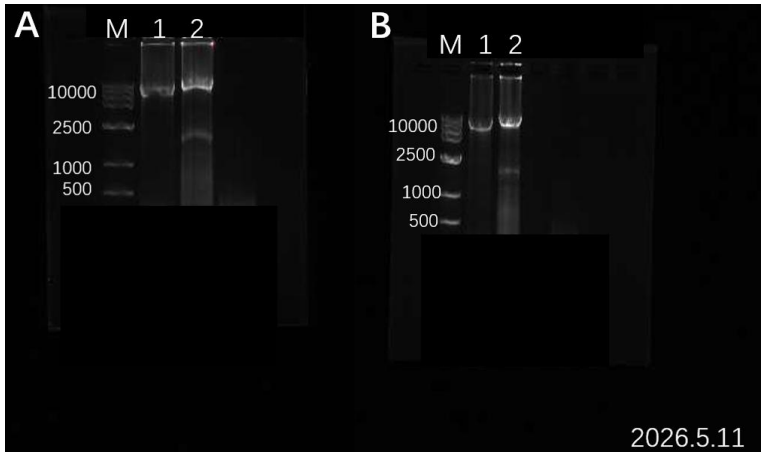
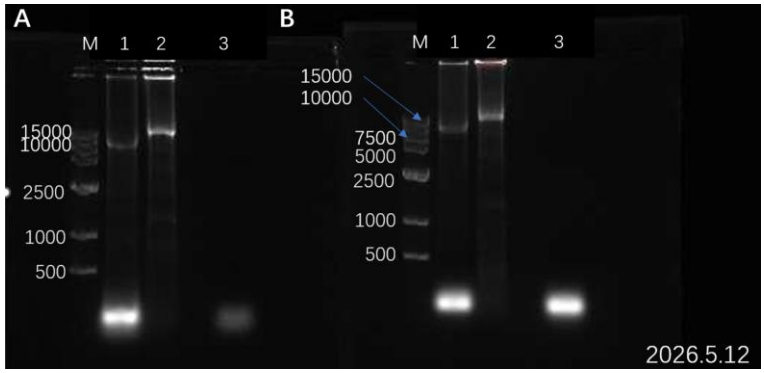
Simple	concentration（ng/ul）	Purity（A260/280）
Z1	58.700	1.950
Z2	69.900	1.910
Z3	64.750	1.939
S1	66.250	1.960
S2	46.400	1.962
S3	60.250	1.956

Concentration and Purity of the pML104 Plasmid

Explanation	The pML104 plasmid was successfully extracted, and its concentration and purity were both found to be within the normal range.
Next plan	Linearize the pML104 plasmid by inverse PCR.

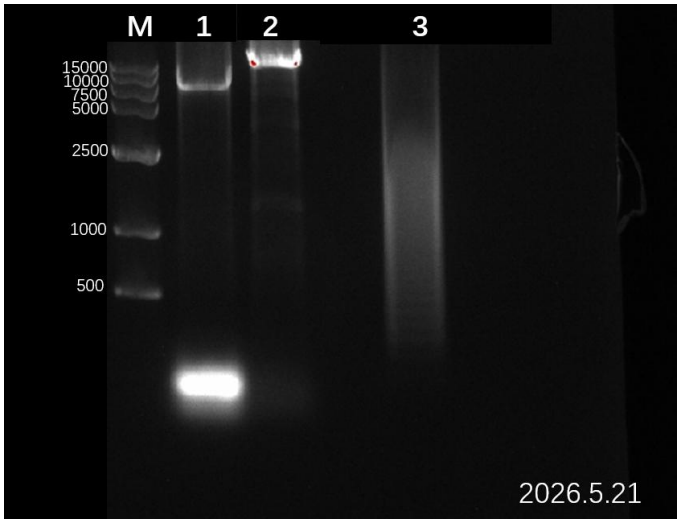
No 3 Date：May 7–12, 2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

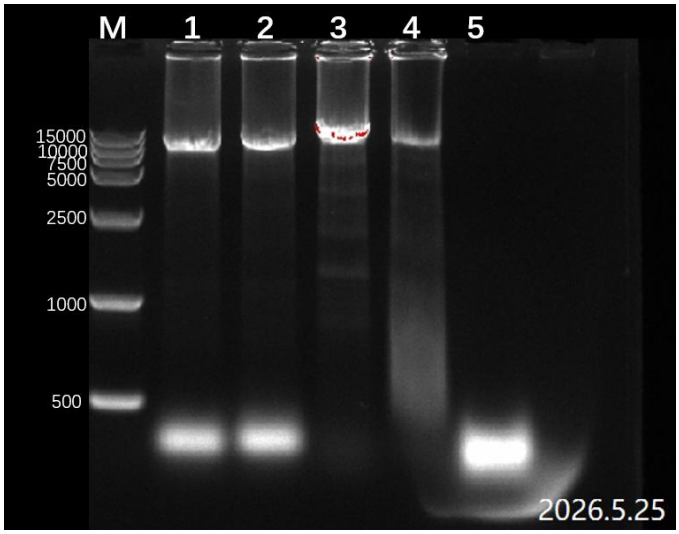
Experiment	Linearization of the pML104 Plasmid by Inverse PCR
Note	Protocol 5:Agarose Gel Electrophoresis ;Protocol 7: Conventional PCR;

Result	 Figure 1. Agarose gel electrophoresis of linearized pML104 obtained by inverse PCR on May 7 and May 8. (A) May 7; (B) May 8. M: DNA marker; lanes 1–6: 48, 51, 53, 55, 57, 59 °C.  **Figure 2. Agarose gel electrophoresis of linearized pML104 obtained by inverse PCR on May 11.** (A)(B) Same procedure. M: DNA marker; 1: undiluted primer; 2: primer working solution.  **Figure 3. Agarose gel electrophoresis of linearized pML104 obtained by inverse PCR on May 12.** (A)(B) Same procedure. M: DNA marker; 1: undiluted primer; 2: diluted primer; 3: negative control.
Explanation	This experiment aimed to linearize the pML104 plasmid by inverse PCR, but has not yet yielded the expected results. An initial annealing temperature gradient

	(48–59 °C) showed no significant differences, so subsequent experiments used a uniform 55 °C based on the primer Tm (May 11 and May 12). Since a band had previously been obtained with undiluted primer, both primer stock and working solution were tested later; however, the working solution bands were consistently positioned higher than expected, and all bands were relatively faint with low yield. In summary, the inverse PCR linearization conditions have not yet been successfully optimized.
Next plan	Optimize the inverse PCR conditions to obtain the expected linearized pML104 product.

No 4 Date：May 21–25, 2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

Experiment	Optimization of PCR Conditions and Repeated Linearization of the pML104 Plasmid
Note	Protocol 5:Agarose Gel Electrophoresis ;Protocol 7: Conventional PCR
Result	 **Figure 1. Agarose gel electrophoresis of linearized pML104 obtained by inverse PCR on May 21.** M: DNA marker; lane 1: undiluted primer; lane 2: diluted primer; lane 3: negative control.

	 **Figure 2. Agarose gel electrophoresis of linearized pML104 obtained by inverse PCR on May 25.** M: DNA marker; lane 1: 55 °C, undiluted primer; lane 2: 53 °C, undiluted primer; lane 3: 55 °C, diluted primer; lane 4: 53 °C, diluted primer; lane 5: negative control.
Explanation	In this experiment, the effect of primer dilution on inverse PCR linearization was first tested. The results showed that although the undiluted primer produced a band of the expected size, direct use of primer stock was considered insufficiently justified and therefore not adopted, whereas the diluted primer yielded a band positioned noticeably higher than expected (Figure 1), indicating that the amplification program still required optimization. On this basis, annealing temperatures of 53 °C and 55 °C were further compared (with extension time extended to 8 min), and no obvious difference was observed between the two conditions (Figure 2), preliminarily ruling out annealing temperature as a major cause.
Next plan	Further optimize the PCR reaction program to improve amplification efficiency.

No 5 Date：July 9, 2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

Experiment	Further Optimization of PCR Conditions for pML104 Linearization
Note	Protocol 5:Agarose Gel Electrophoresis ;Protocol 7: Conventional PCR; Protocol 9: PCR Purification

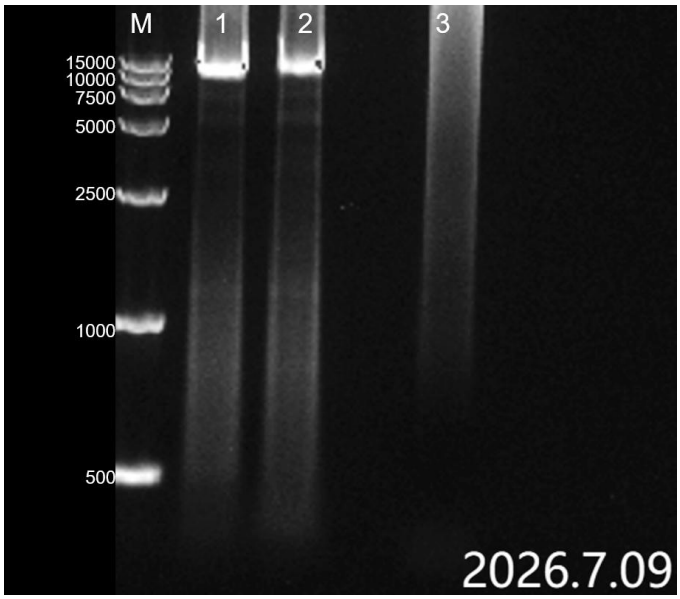
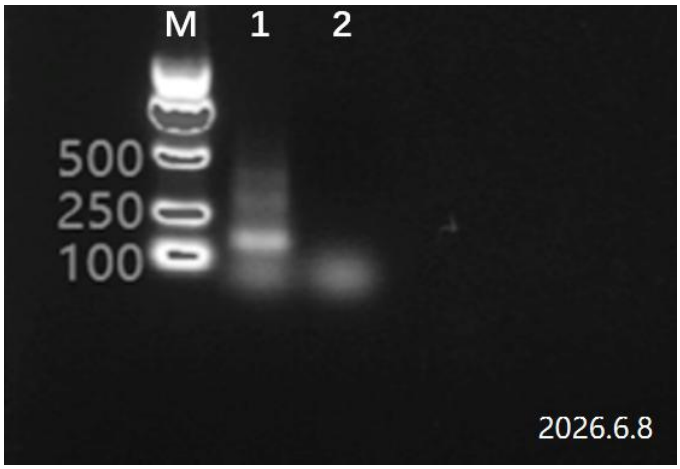
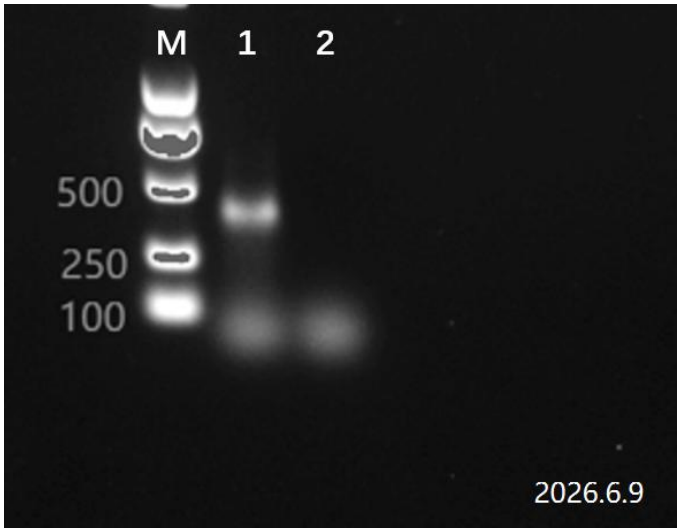
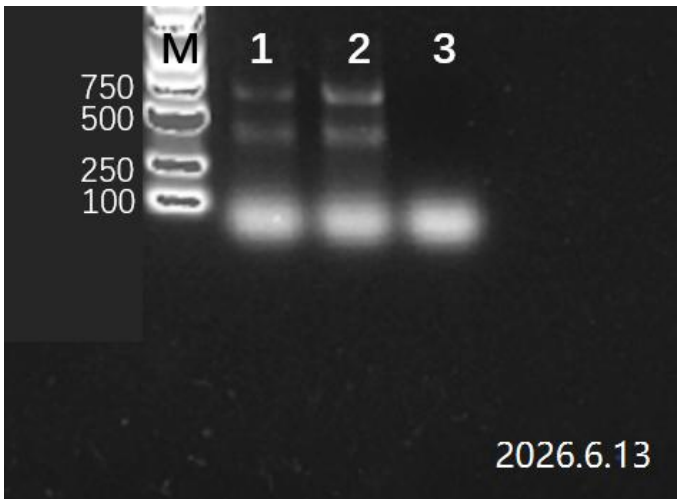
<table>
<tr><td>Result</td><td>


Figure 1. Agarose gel electrophoresis of linearized pML104 obtained by inverse PCR on July 9.
M: DNA marker; lane 1: annealing temperature 53 °C; lane 2: annealing temperature 55 °C; lane 3: negative control.
<table>
<tr><th>Simple</th><th>Concentration（ng/ul）</th><th>Purity（A260/280）</th></tr>
<tr><td>1(Annealing temperature: 53℃)</td><td>242.75</td><td>1.855</td></tr>
<tr><td>2(Annealing temperature: 55℃)</td><td>69.500</td><td>1.812</td></tr>
</table>
Concentration and Purity of the Gel Extraction Product (July 9, 2026)
</td></tr>
<tr><td>Explanation</td><td>The results showed a distinct band between 10000 bp and 15000 bp, preliminarily identified as the target band (11209 bp). In this experiment, an optimized program with an extension temperature of 65 °C was applied to the 11.2 kb linearized vector containing AT-rich repeats, and specific amplification was successfully achieved, indicating that extension temperature optimization was the key factor in overcoming the difficulty of amplifying AT-rich templates.</td></tr>
<tr><td>Next plan</td><td>Since the pML104 vector has been successfully linearized, it will next be used for one-step cloning and transformed into E. coli to obtain the correct recombinant plasmid. The correct recombinant plasmid will then be used for CRISPR-Cas9 gene knockout in Saccharomyces cerevisiae to construct FAR1 and SST2 knockout strains.</td></tr>
</table>

No 6 Date：June 8–July 14,2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

Experiment	One-Step Cloning of the Linearized pML104 Vector and E. coli Transformation
Note	Protocol 1:E. coli Transformation and Culture Expansion ;Protocol 5:Agarose Gel Electrophoresis ;Protocol 7: Conventional PCR; Protocol 11: Gibson Assembly;

Result	 **Figure 1. PCR verification of the one-step cloning product on June 8.** M: DNA marker; lane 1: one-step cloning product; lane 2: negative control (water instead of template).  **Figure 2. PCR verification of the one-step cloning product on June 9.** M: DNA marker; lane 1: one-step cloning product; lane 2: negative control.  **Figure 3. PCR verification of one-step cloning products with different reaction volumes on June 13.**

M: DNA marker; lane 1: 20 μL one-step cloning reaction; lane 2: 10 μL one-step cloning reaction; lane 3: negative control.

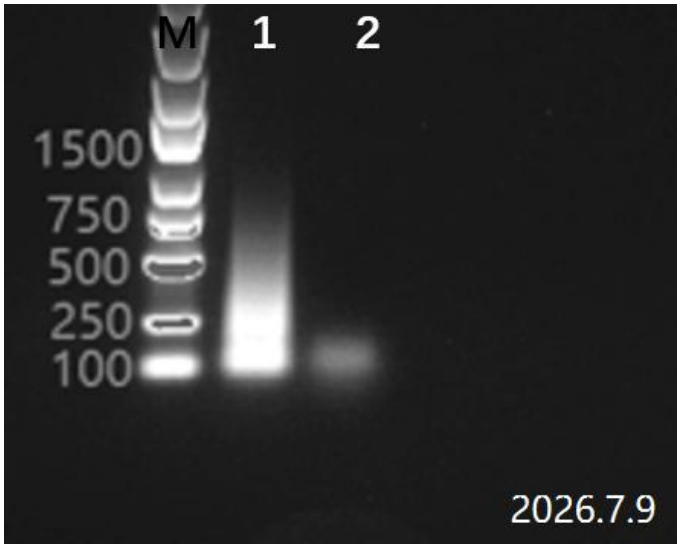


Figure 4. PCR verification of the one-step cloning product on July 9.
M: DNA marker; lane 1: one-step cloning product; lane 2: negative control.

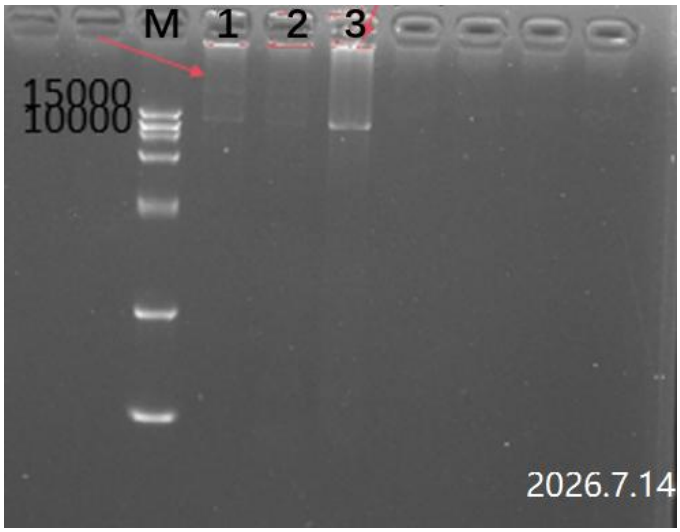


Figure 5. Agarose gel electrophoresis comparison of one-step cloning products and the pML104 empty plasmid.
M: DNA marker; lane 1: one-step cloning product from July 10; lane 2: one-step cloning product from July 14; lane 3: pML104 empty plasmid (non-linearized).

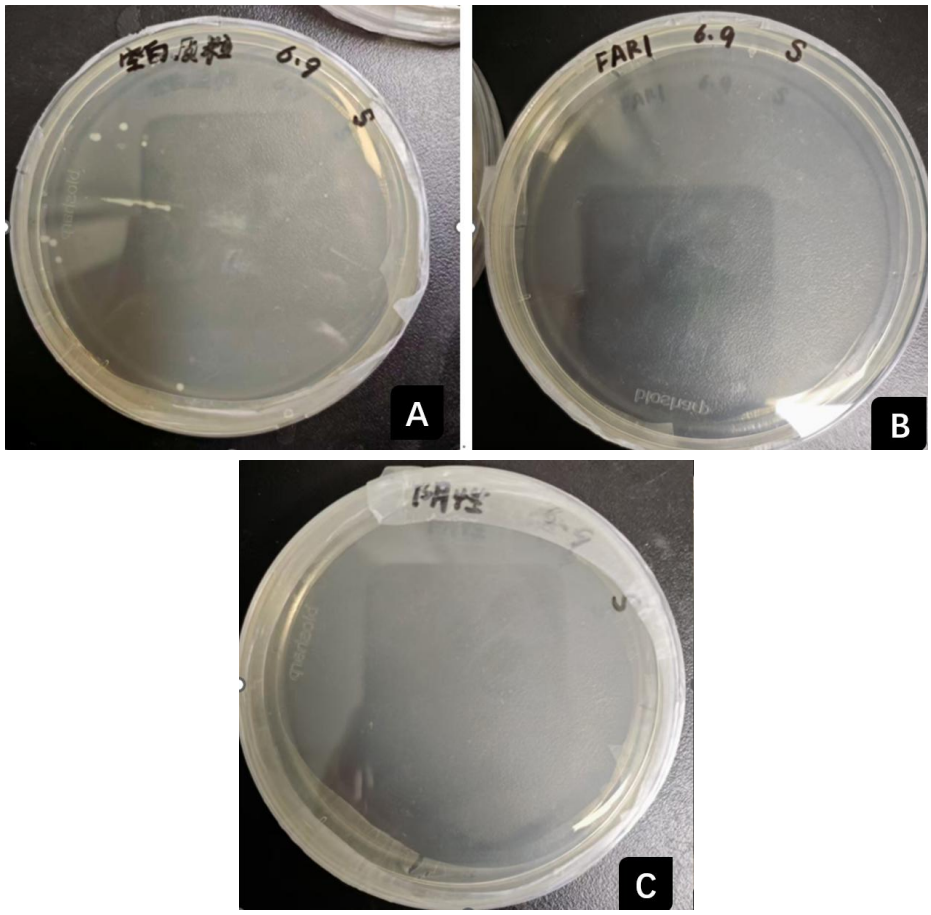
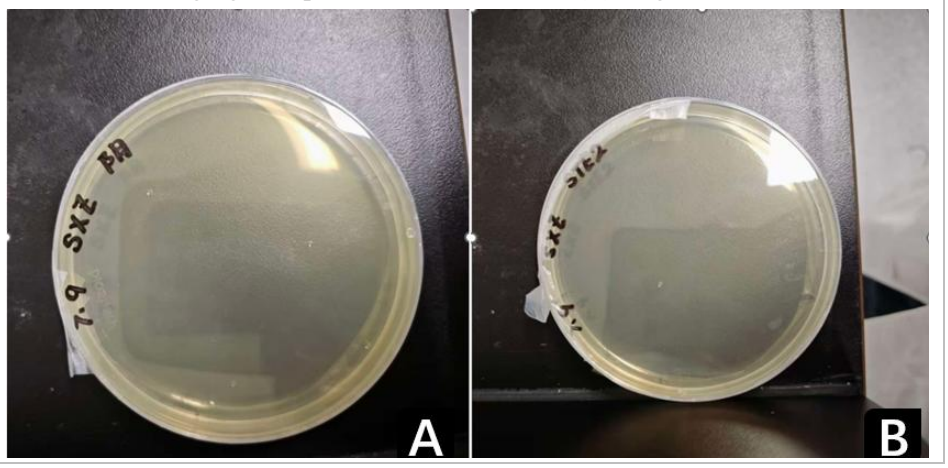


Figure 6. Plate results of E. coli Top10 transformation with the one-step cloning product on June 9.

(A) pML104 empty plasmid transformation (positive control); (B) One-step cloning ligation product transformation; (C) Negative control.



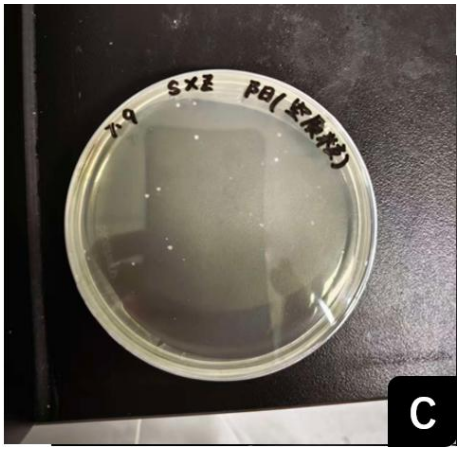


Figure 7. Plate results of E. coli Top10 transformation with the one-step cloning product on July 9.

(A) Negative control; (B) One-step cloning ligation product transformation; (C) pML104 empty plasmid transformation (positive control). All were plated on LB agar with the corresponding antibiotic.

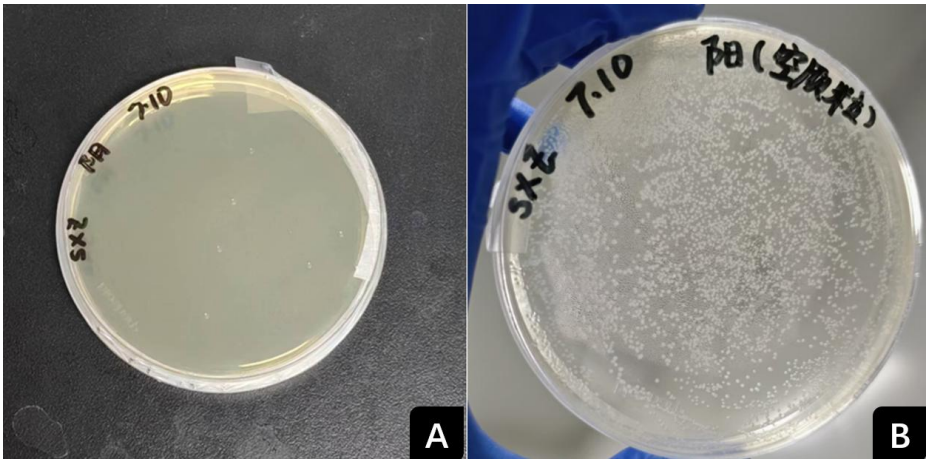




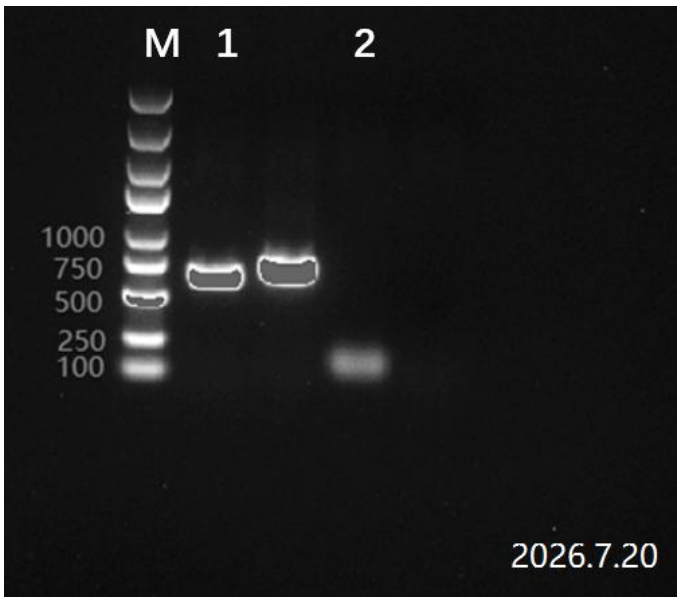
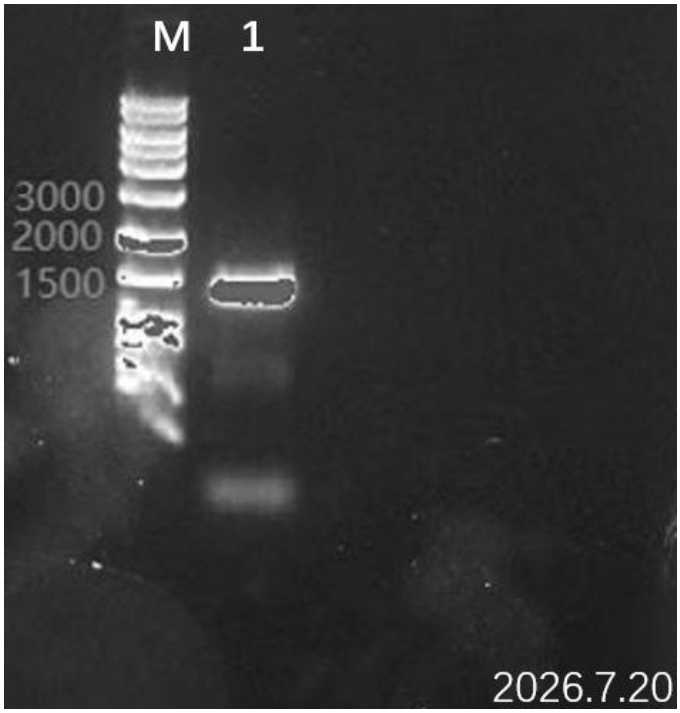
Figure 8. Plate results of E. coli Top10 transformation with the one-step cloning product on July 10.

(A) Negative control; (B) pML104 empty plasmid transformation (positive control); (C) One-step cloning ligation product transformation. All were plated on LB agar with the corresponding antibiotic.

Explanation	The one-step cloning experiment was first verified by PCR. Early results showed excessive non-specific amplification, making the target band difficult to identify (Figure 1). In subsequent attempts, a target band of approximately 406 bp was obtained, with a position consistent with the expected size and a bright, clear appearance (Figure 2); however, subsequent E. coli transformation did not yield successful transformants. When different reaction systems were compared, target bands were observed in both systems, but each was accompanied by a relatively obvious non-specific band (Figure 3). After replacing the linearized vector with a new batch (concentration increased to 69 ng/μL) and performing ligation at a molar ratio of insert to vector of 1:10, PCR verification still failed to produce the correct target band (Figure 4). To preliminarily assess whether the size of the cloning product was correct, the one-step cloning products and the empty plasmid were directly loaded for agarose gel electrophoresis comparison (Figure 5). Both the July 10 and July 14 one-step cloning products showed bands between 10000 bp and 15000 bp, with an additional control containing 2 μL of linearized vector. Since open-circular plasmids migrate more slowly and appear higher on the gel, the more prominent band may correspond to the unligated linearized vector, with a position roughly consistent with the expected size. Transformation experiments further showed that the empty plasmid group produced only a few scattered colonies (possibly due to excessive plasmid concentration), the negative control group showed no colonies, and the one-step cloning product group also produced no colonies, suggesting that the insert and vector may not have been successfully ligated (Figure 6). The transformation procedure was then modified by adding a centrifugation step to collect the cells, followed by resuspension in 100 μL of medium before plating. The results showed no colonies in either the negative control or the one-step cloning ligation product group, while the empty plasmid positive control showed colony growth, further suggesting that the insert and vector were still not successfully ligated (Figure 7). After further adjusting the volume of ligation product (from 10 μL to 9 μL) while retaining the resuspension step, the positive control produced numerous colonies and the negative control produced none, indicating that the transformation procedure itself was not problematic and that the issue lay in the failure of the insert and vector to ligate successfully (Figure 8).
Next plan	Since the one-step cloning failed, we plan to redesign the linearization strategy by introducing the target fragment directly through primers, so that the linearized fragment produced by inverse PCR contains both the vector and the target insert, thereby bypassing the ligation step.

No 7 Date：July 20–22, 2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

Experiment	PCR Amplification of Repair Templates and Target Gene Fragments for FAR1 and SST2 Gene Knockout

Note	Protocol 5:Agarose Gel Electrophoresis ; Protocol 7:Conventional PCR ;Protocol 13:Colony PCR for Saccharomyces cerevisiae;
Result	 **Figure 1. PCR amplification of repair templates for FAR1 and SST2 knockout on July 20.** M: DNA marker; lane 1: repair template for FAR1 (611 bp); lane 2: repair template for SST2 (1303 bp).  **Figure 2. Amplification of the SST2 repair template after PCR optimization on July 20.** M: DNA marker; lane 1: SST2 repair template PCR product(1303bp).

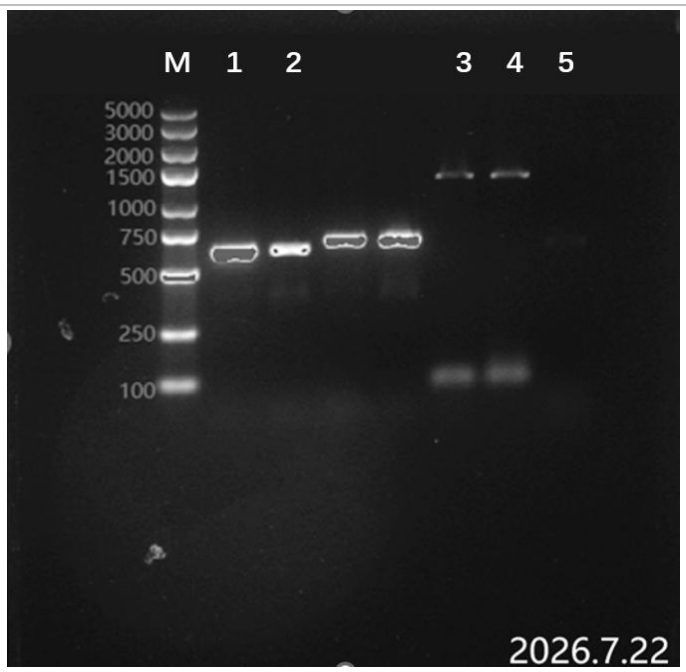


Figure 3. PCR amplification of FAR1 and SST2 repair templates on July 22.
M: DNA marker; lanes 1–2: FAR1 repair template PCR products; lanes 3–4: SST2 repair template PCR products; lane 5: negative control.

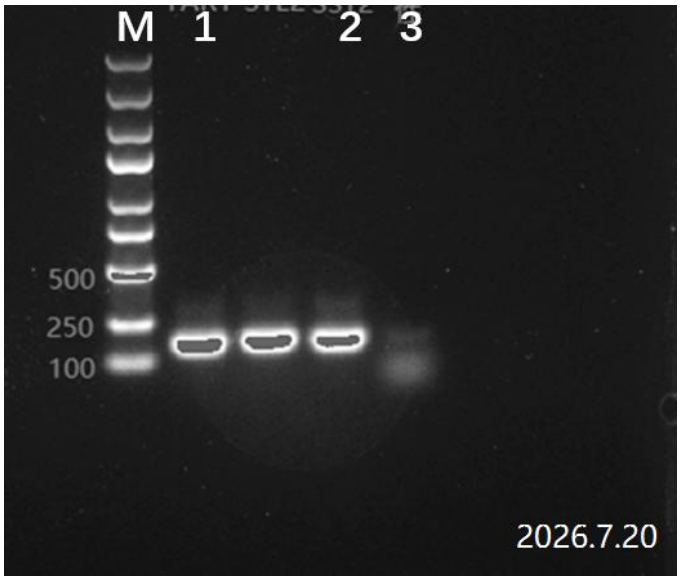
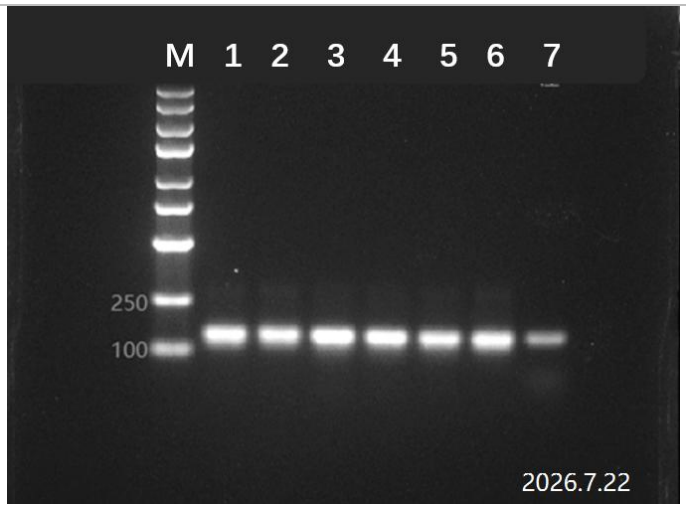
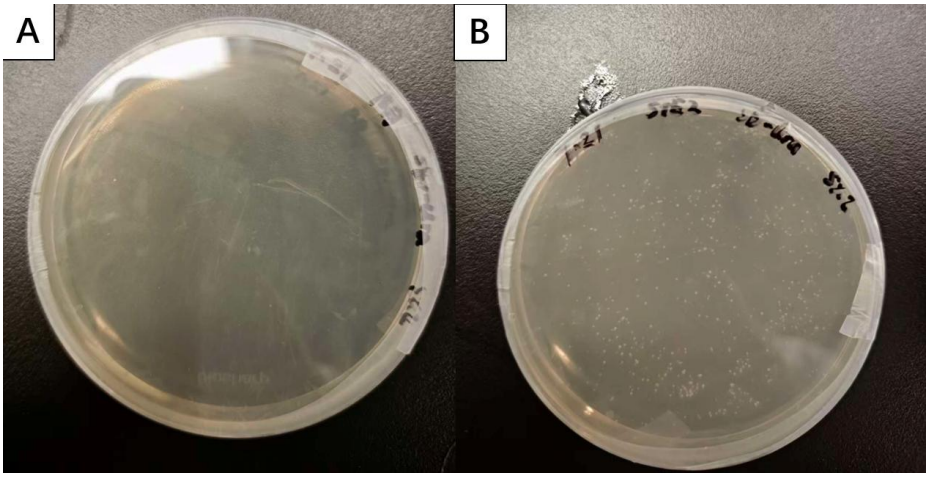
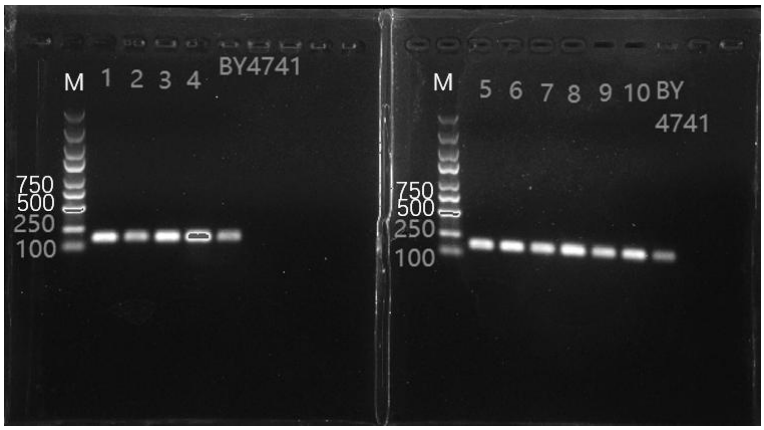


Figure 4. PCR amplification of target gene fragments on July 20.
M: DNA marker; lane 1: Far1 target gene fragment; lane 2: Sst2 target gene fragment; lane 3: negative control.

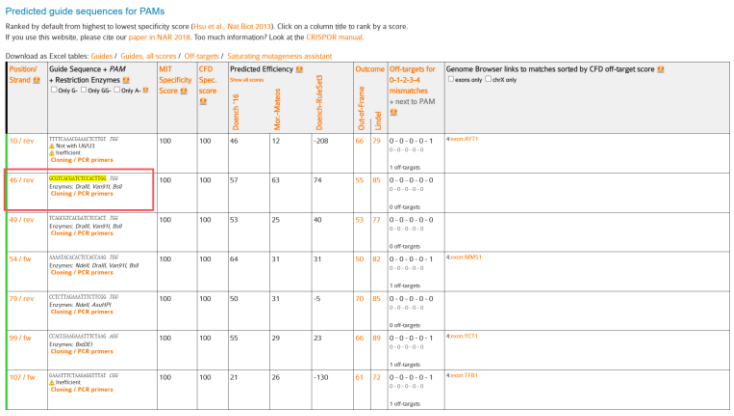
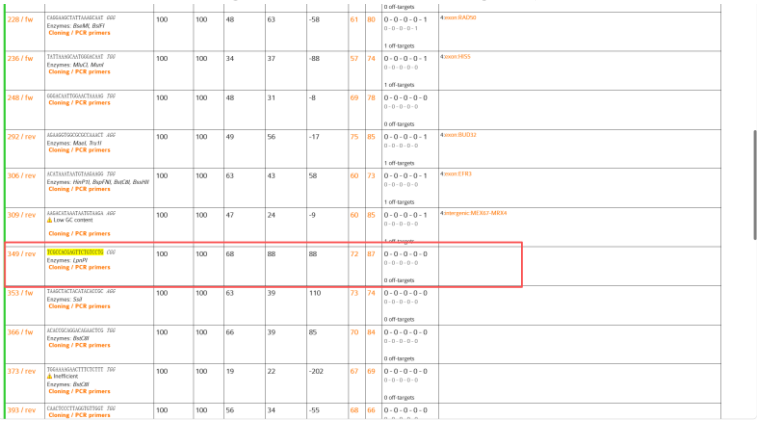
	 **Figure 5. PCR amplification of target gene fragments on July 22.** M: DNA marker; lanes 1–3: FAR1 target gene fragment (128 bp); lanes 4–6: SST2 target gene fragment; lane 7: negative control.
Explanation	In this experiment, PCR amplification was first performed to obtain the repair templates required for FAR1 and SST2 knockout. The expected sizes were 611 bp for FAR1 and 1303 bp for SST2. The results showed a clear band of the correct size for FAR1, whereas no band was observed for SST2 (Figure 1), possibly due to insufficient extension time. After extending the extension time for SST2 from 30 s to 1 min, a clear band of the correct size (1303 bp) was successfully obtained, confirming that the previous failure was caused by insufficient extension time (Figure 2). When the reaction system was scaled up for recovery, the FAR1 band was clear and of the expected size, whereas the SST2 band was positioned higher than expected, clearly larger than the theoretical 1303 bp and even above the 1500 bp marker band, suggesting it was not the target band; therefore, only the FAR1 product was recovered (Figure 3). Subsequently, PCR amplification of the target gene fragments was performed. A contaminating band was observed in the negative control lane, suggesting background interference such as non-specific amplification or primer dimers; however, the target bands were clear and single, with no other non-specific products, indicating good amplification specificity and feasibility of direct purification (Figure 4). On this basis, the reaction system was further scaled up to increase the recovery yield (Figure 5). These results indicate that, after optimizing the extension time, both the FAR1 and SST2 repair templates and target gene fragments were successfully amplified, providing materials for subsequent one-step cloning and knockout vector construction.
Next plan	Since the inverse PCR linearization of pML104 has not yet been successful, we will continue to optimize the linearization PCR to obtain the correct pML104-sgRNA-Far1 and pML104-sgRNA-Sst2 constructs. Once the plasmids are correctly constructed, they will be co-transformed into yeast for subsequent gene knockout experiments.

No 8 Date：July 21–23,2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

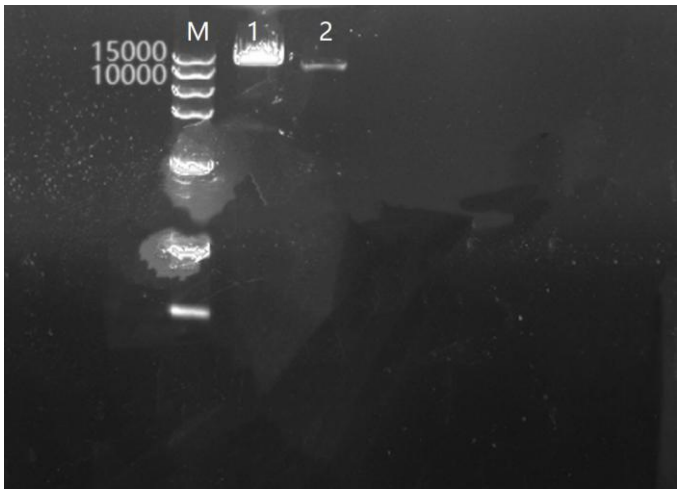
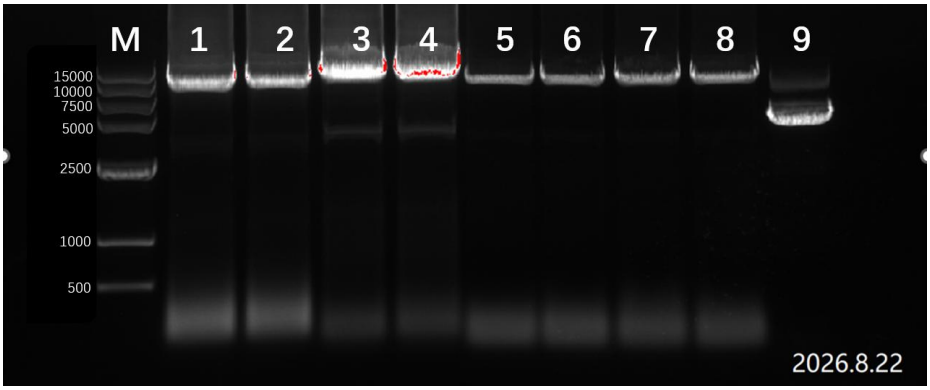
Experiment	Co-Transformation of the Linearized Vector, Linearized Insert, and Repair Templates into Saccharomyces cerevisiae and Colony PCR Verification
Note	Protocol 5:Agarose Gel Electrophoresis;Protocol 12: Yeast Transformation ;Protocol 13:Colony PCR for Saccharomyces cerevisiae; Protocol 19: CRISPR-Cas9 Transformation for Gene Knockout in Yeast;
Result	 **Figure 1. Plate results of co-transformation of the linearized vector, linearized insert, and repair templates into Saccharomyces cerevisiae.** (A) Negative control; (B) Transformants obtained after co-transformation with the linearized vector, linearized insert, and repair templates. Both were plated on SD-His-Ura selective medium.  **Figure 2. Colony PCR screening of transformants obtained by co-transformation of the linearized vector, linearized insert, and repair templates.** M: DNA marker; lanes 1–10: co-transformants; BY / BY4741: wild-type control.
Explanation	To test whether co-transformation of the linearized vector, linearized insert, and repair templates could be used for Far1 and Sst2 knockout, the linearized pML104 vector, the linearized insert containing the guide sequence, and the repair templates were co-transformed into Saccharomyces cerevisiae BY4741 and plated on SD-His-Ura selective medium. Single colonies were obtained on the plate (Figure 1), and 10 single colonies were picked for colony PCR screening, with wild-type BY4741 as the control. Since the primers were designed at the

	knockout site, successful gene knockout prevents primer binding and therefore no target band is amplified, whereas wild-type BY4741 amplifies normally. The results showed that both wild-type BY4741 and all co-transformants produced clear bands between 100 bp and 250 bp at the same position, with no band pattern consistent with successful knockout (Figure 2), indicating that Far1 and Sst2 were not successfully knocked out. Therefore, Sanger sequencing was not performed.
Next plan	The linearization strategy and repair templates will be redesigned in subsequent experiments to improve the efficiency of CRISPR-Cas9-mediated knockout in yeast.

No 9 Date：August 6, 2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

Experiment	Redesign and Construction Strategy of sgRNA Target Sites
Note	sgRNA target sites were redesigned using CRISPOR (http://crispor.tefor.net/).
Result	 Predicted guide sequences for PAMs Ranked by default from highest to lowest specificity score (Hsu et al., Nat Biot 2013). Click on a column title to rank by a score. If you use this website, please cite our paper in NAR 2018. Too much information? Look at the CRISPOR manual. Download as Excel tables: Guides / Guides, all scores / Off-targets / Saturating mutagenesis assistant **SgRNA of Far1(redesigned)**  **SgRNA of Sst2(redesigned)**
Explanation	The sgRNA target sites were redesigned using CRISPOR (http://crispor.tefor.net/). Selection was based on a combination of the MIT specificity score, CFD cutting efficiency score, and the number of off-target sites. Ultimately, new target sites with high specificity (MIT > 90), relatively high CFD scores, and zero off-target sites were selected for subsequent construction.
Next plan	New primers will be designed based on the redesigned sgRNA target sites for linearization of the vector.

No 10 Date：August 21–22, 2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

Experiment	Redesigned Inverse PCR Linearization of pML104 for Direct Insert Introduction
Note	Protocol 5:Agarose Gel Electrophoresis ;Protocol 7:Conventional PCR; Protocol 11: Gibson Assembly
Result	 **Figure 1. Agarose gel electrophoresis of inverse PCR linearization of the pML104-sgRNA-Far1 and pML104-sgRNA-Sst2 constructs on August 21.** M: DNA marker; lane 1: linearized pML104-sgRNA-Far1 product(11258bp); lane 2: linearized pML104-sgRNA-Sst2 product(11258bp).  **Figure 2. Agarose gel electrophoresis of inverse PCR linearization of the pML104-sgRNA-Far1 and pML104-sgRNA-Sst2 constructs on August 22.** M: DNA marker; lanes 1–4: linearized pML104-sgRNA-Far1 products(11258bp); lanes 5–8: linearized pML104-sgRNA-Sst2 products(11258bp); lane 9: pML104 empty plasmid (circular, non-linearized).

Sample	Concentration（ng/ul）	Purity（A260/280）
pML104-sgRNA-Far1(200mg)	43.600	1.900
pML104-sgRNA-Sst2(310mg)	33.400	1.925

Concentration and purity of the linearized pML104-sgRNA-Far1 and pML104-sgRNA-Sst2 products.

Explanation	Since the target fragment (sgRNA) is very short, a redesigned strategy was adopted: the pML104 empty plasmid was used directly as the template, and the target fragment was introduced through primer design during inverse PCR amplification, so that the resulting linearized fragment contains both the vector backbone and the target insert. The theoretical length of the amplified product is approximately 11258 bp, and the band position was preliminarily judged to be consistent with the expected size (Figure 1). On this basis, the reaction system was further scaled up to increase product yield. In Figure 2, lanes 1–4 correspond to the linearized pML104-sgRNA-Far1 product, and lanes 5–8 correspond to the linearized pML104-sgRNA-Sst2 product. Although all four lanes were Far1 samples, the bands in lanes 1 and 2 were positioned noticeably lower than those in lanes 3 and 4. Since the theoretical size of the target band is 11258 bp, its position should be closer to the 10000 bp marker than to the 15000 bp marker; therefore, the bands in lanes 1 and 2 were judged to be consistent with the expected size, and only lanes 1 and 2 were excised for subsequent recovery, while lanes 3 and 4 were not excised. The pML104-sgRNA-Sst2 bands in lanes 5–8 showed similar positions and were consistent with the expected size, so all of them were excised and recovered for the next experiment. These results indicate that the redesigned inverse PCR linearization strategy is feasible and that both the pML104-sgRNA-Far1 and pML104-sgRNA-Sst2 linearized products were successfully obtained, allowing subsequent transformation and colony PCR verification.
Next plan	The linearized pML104-sgRNA-Far1 and pML104-sgRNA-Sst2 products will be used for one-step cloning and transformed into E. coli to obtain the correct recombinant plasmids.

No 11 Date：August 20, 2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

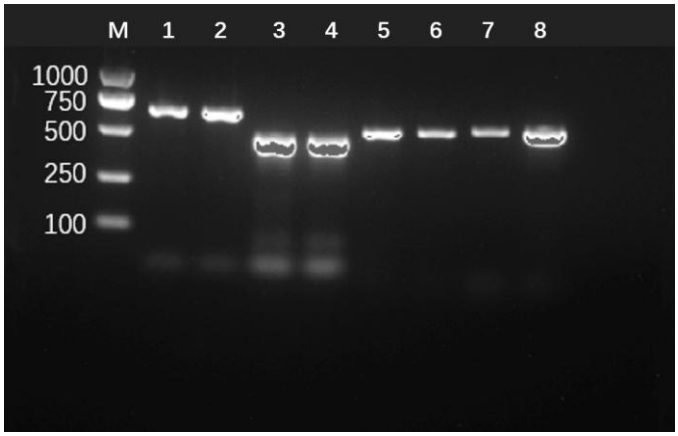
Experiment	PCR Amplification of Redesigned Repair Templates for FAR1 and SST2 Knockout
Note	Protocol 5:Agarose Gel Electrophoresis ;Protocol 8 :Overlap Extension PCR; Protocol 9: PCR Purification; Protocol 10: Gel Extraction of PCR Products; Protocol 13:Colony PCR for Saccharomyces cerevisiae
Result	

Figure 1. PCR amplification of the upstream and downstream homology arms for the redesigned FAR1 and SST2 knockout repair templates.

M: DNA marker; lanes 1–2: FAR1 upstream homology arm(658bp); lanes 3–4: SST2 upstream homology arm(491bp); lanes 5–6: FAR1 downstream homology arm; lanes(528bp) 7–8: SST2 downstream homology arm(542bp).

Sample	Concentration（ng/ul）	Purity(A260/280)
Far1 upstream arm(510mg)	78.5	1.874
Sst2 upstream arm(520mg)	31.45	1.878
Far1 downstream arm	159.8	1.870
Sst2 downstream arm(660mg)	99.70	1.883

Concentration and purity of the recovered upstream and downstream homology arms for the FAR1 and SST2 repair templates.

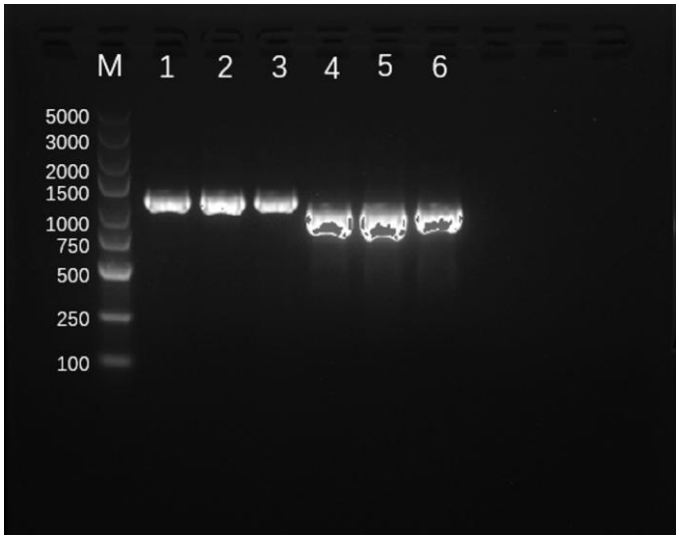
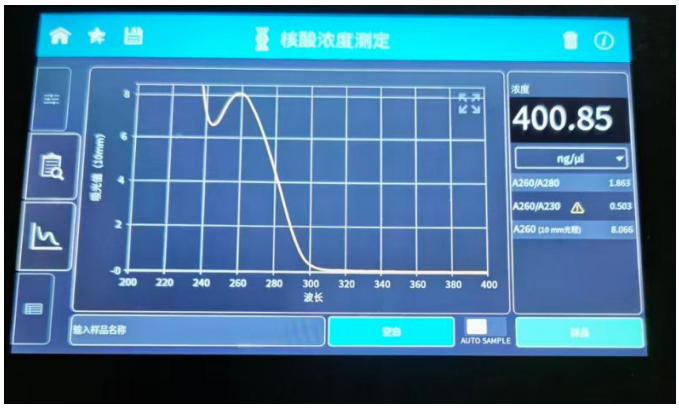


Figure 2. PCR amplification of repair templates for FAR1 and SST2 genes after fusion PCR assembly.

M: DNA marker; lanes 1–3: repair template for the FAR1 gene (1146 bp); lanes 4–6: repair template for the SST2 gene (993 bp).







Sample	Concentration（ng/ul）	Purity(A260/280)
Repair template for the Far1 gene	174.95	1.860
Repair template for the Sst2 gene	365.20	1.868
Repair template for the Sst2 gene	400.85	1.863

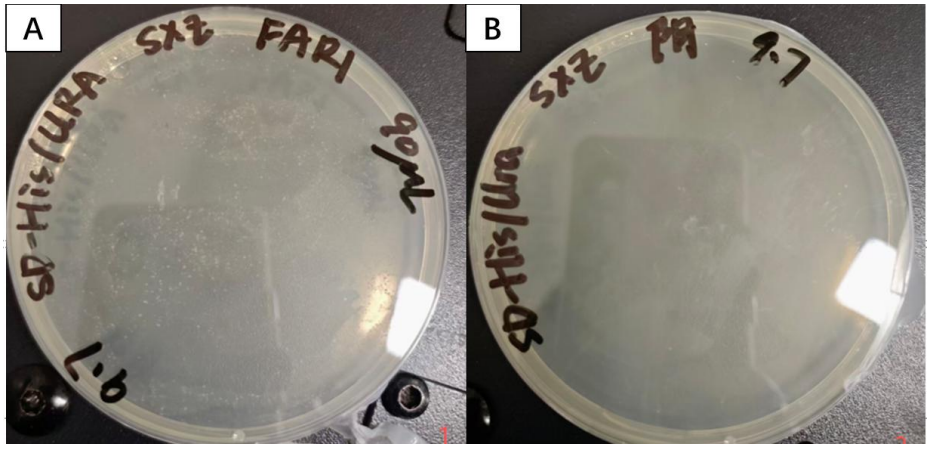
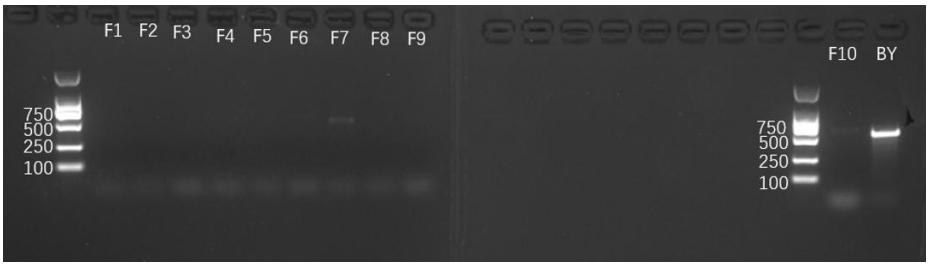
Concentration and purity of the purified repair templates after fusion PCR assembly.

Explanation

Since the previous repair template design did not meet the requirements of subsequent experiments, the knockout repair templates for FAR1 and SST2 were redesigned: approximately 500 bp homology arms were designed upstream and downstream of the knockout site, and fusion PCR was used to assemble them into the full-length repair template, allowing the target mutation to be introduced through homologous recombination after CRISPR-Cas9 cleavage. To obtain these templates, the upstream and downstream homology arms of FAR1 and SST2 were amplified separately, and the recovery method was chosen based on the electrophoresis results: the FAR1 downstream arm showed a clear band and was recovered by direct purification, while the other three groups showed non-specific bands and were recovered by gel extraction (Figure 1). After fusion PCR assembly, both the FAR1 and SST2 repair templates showed clear bands of the expected sizes (1146 bp for FAR1 and 993 bp for SST2), so the assembled products were recovered directly (Figure 2). These results indicate that the redesigned repair templates for the FAR1 and SST2 genes were successfully

	amplified and assembled, and can be used for subsequent CRISPR-Cas9 co-transformation and gene knockout experiments.
Next plan	The purified repair templates will be co-transformed into Saccharomyces cerevisiae together with the constructed pML104-sgRNA-Far1 and pML104-sgRNA-Sst2 plasmids for FAR1 and SST2 gene knockout.

No 12 Date：September 7–20, 2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

Experiment	Co-Transformation of Linearized pML104-sgRNA Plasmids and Repair Templates into Saccharomyces cerevisiae and Verification
Note	Protocol 5:Agarose Gel Electrophoresis;Protocol 12: Yeast Transformation ;Protocol 13:Colony PCR for Saccharomyces cerevisiae; Protocol 19: CRISPR-Cas9 Transformation for Gene Knockout in Yeast;
Result	 **Figure 1. Plate results of co-transformation of the linearized pML104-sgRNA-Far1 plasmid and the FAR1 repair template into Saccharomyces cerevisiae.** (A) Co-transformation group; (B) Negative control.  **Figure 2. Colony PCR screening of FAR1 knockout transformants.** M: DNA marker; F1–F9: FAR1 knockout transformants; F10: negative control; BY: wild-type BY4741 control.

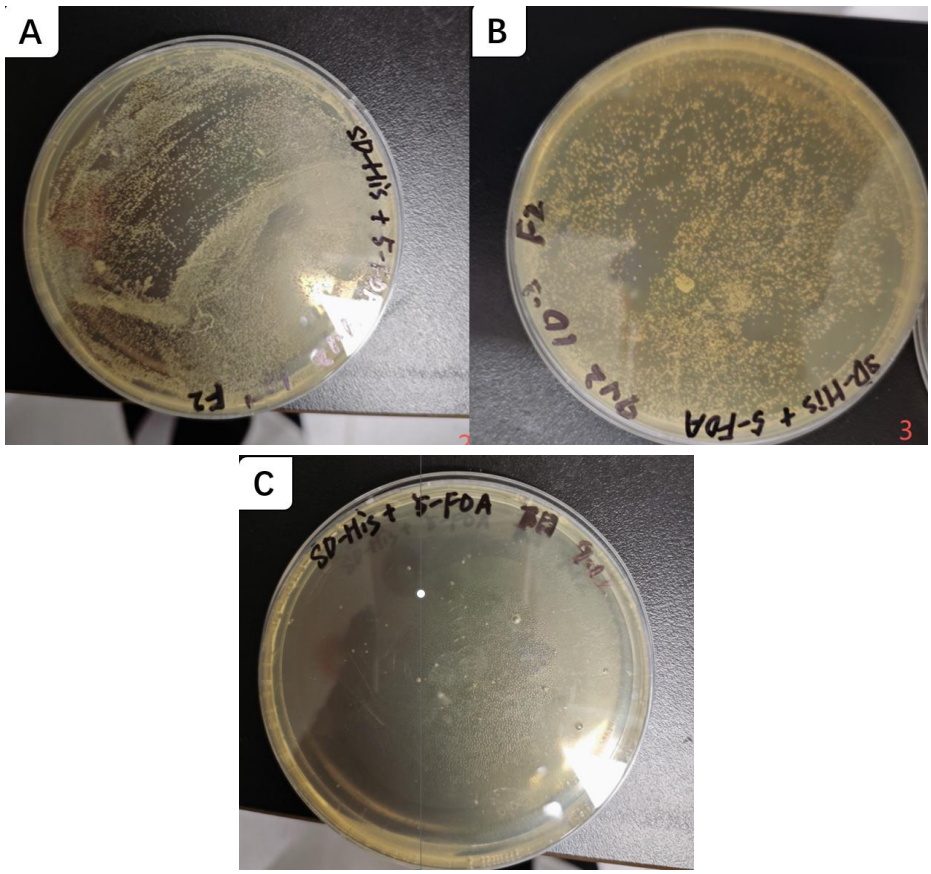


Figure 3. Plate results of 5-FOA counter-selection for plasmid removal.
(A) Transformants diluted to 10^{-1}; (B) Transformants diluted to 10^{-2}; (C) Negative control. All were plated on SD-His + 5-FOA selective medium.

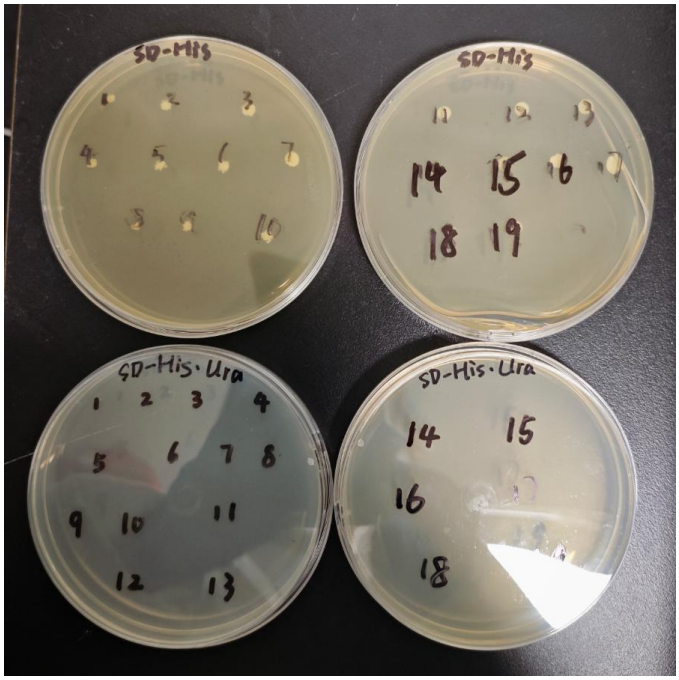
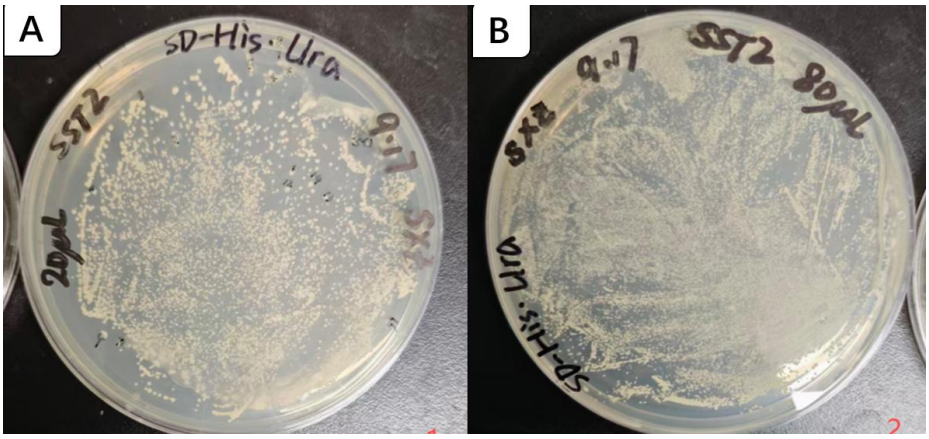
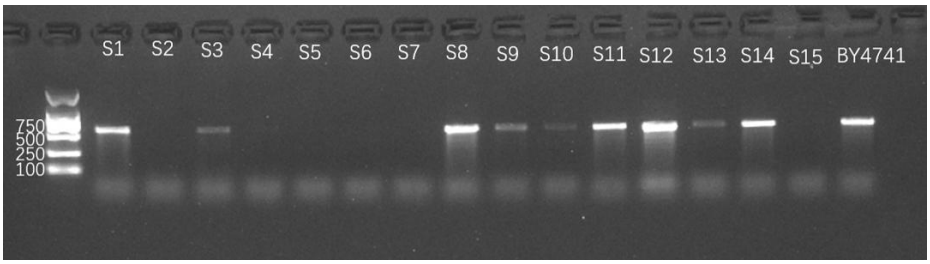
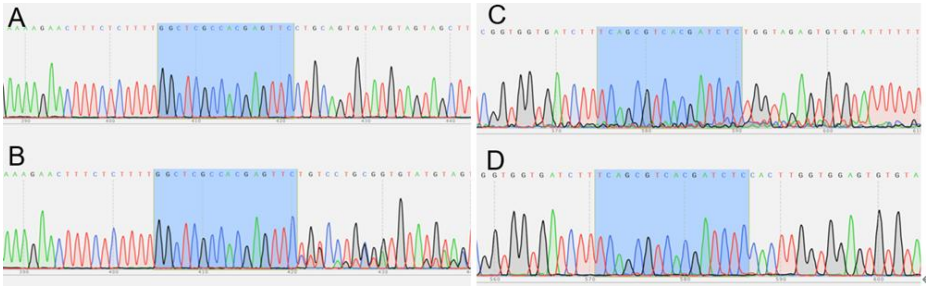


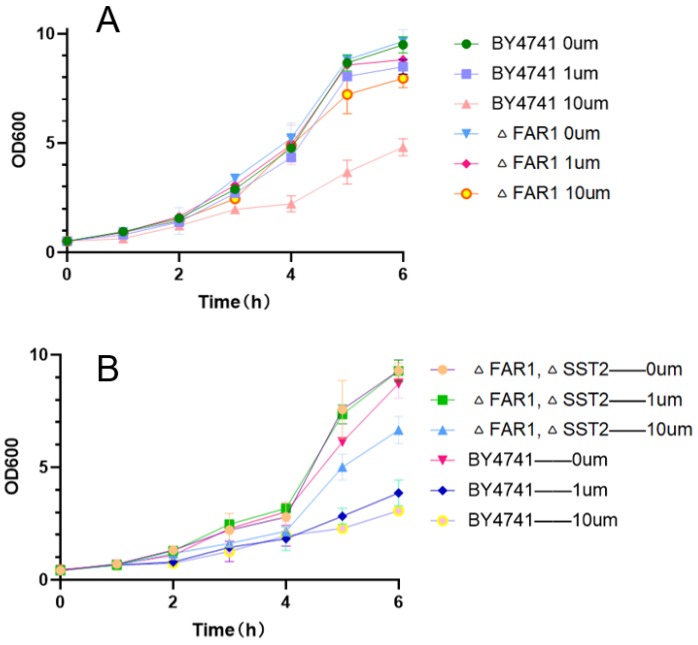
Figure 4. Spot assay of single colonies for plasmid loss after 5-FOA counter-selection.
Top row: SD-His plates; bottom row: SD-His-Ura plates. Numbers 1–19 indicate

	individual colonies.  **Figure 5. Plate results of co-transformation of the linearized pML104-sgRNA-SST2 plasmid and the SST2 repair template into Saccharomyces cerevisiae.** (A) 20 μL of transformation mixture plated; (B) 80 μL of transformation mixture plated. Both were plated on SD-His-Ura selective medium.  **Figure 6. Colony PCR screening of FAR1 and SST2 knockout transformants.** M: DNA marker; lanes S1–S15: SST2 knockout transformants; BY4741: wild-type control. Successful knockout prevents primer binding, resulting in no amplified band. Positive clones were S1, S5–S7, and S15.  **Figure 7. Sanger sequencing verification of FAR1 and SST2 knockout.** (A) Sequencing chromatogram of a successfully knocked-out SST2 clone, showing a 4 bp deletion (TGTC); (B) Non-knockout (wild-type) SST2, shown for comparison; (C) Sequencing chromatogram of a successfully knocked-out FAR1 clone, showing a 4 bp deletion (CACT); (D) Non-knockout (wild-type) FAR1, shown for comparison.
Explanation	To construct FAR1 and SST2 knockout strains, a stepwise knockout strategy was applied. First, the linearized pML104-sgRNA-Far1 plasmid and the Far1 repair template were co-transformed into Saccharomyces cerevisiae BY4741

	competent cells and plated on SD-His-Ura selective medium. Single colonies were picked for colony PCR screening: since the primers were designed at the knockout site, successfully knocked-out clones cannot amplify the target band, whereas wild-type BY4741 amplifies normally. Far1 knockout-positive clones (F1–F6, F8, and F9) were thus identified. The residual pML104 plasmid was then removed by 5-FOA counter-selection, and plasmid loss was confirmed by spotting on SD-His and SD-His-Ura plates, yielding a plasmid-free ΔFar1 strain. On this basis, the linearized pML104-sgRNA-Sst2 plasmid and the Sst2 repair template were constructed and co-transformed into the ΔFar1 strain. Sst2 knockout-positive clones (S1, S5–S7, and S15) were identified by colony PCR, and the residual plasmid was again removed by 5-FOA counter-selection. Finally, knockout-positive clones were subjected to Sanger sequencing and aligned with the wild-type sequence: the successfully knocked-out Sst2 clone showed a 4 bp deletion (TGTC), and the successfully knocked-out Far1 clone showed a 4 bp deletion (CACT), while non-knockout clones matched the wild-type sequence. These 4 bp deletions are expected to cause frameshift mutations and disrupt gene function. In summary, ΔFar1 ΔSst2 double-knockout strains were successfully obtained for subsequent functional validation.
Next plan	Functional validation of the ΔFAR1 ΔSST2 strains will be performed. Growth curves will first be measured to assess the effect of α-factor induction on yeast growth, providing a basis for subsequent fluorescence signal detection.

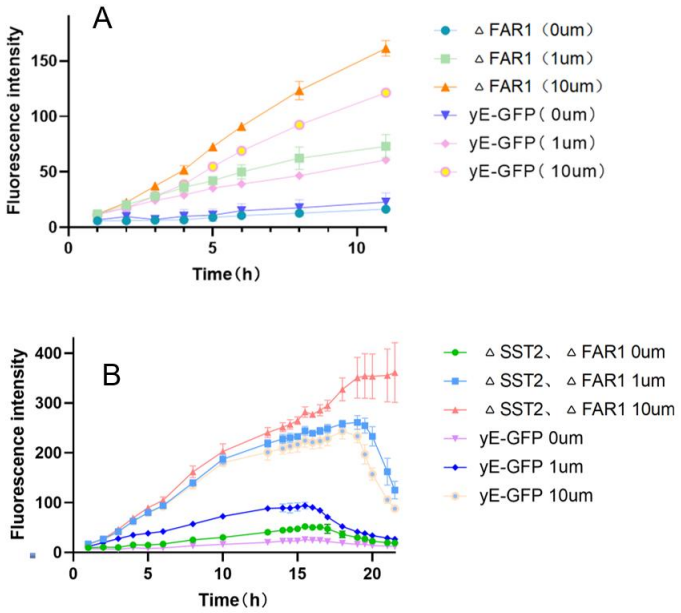
No 13 Date：September 5–14, 2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

Experiment	Functional Validation of ΔFar1 ΔSst2 Strains: Growth Curve Measurement under α-factor Induction
Note	Protocol 14:Yeast Growth Curve Measurement

Result	 **Figure 1. Effect of Far1 and Sst2 knockout on the growth of Saccharomyces cerevisiae.** (A) Growth curves of ΔFar1 and wild-type BY4741 at different α-factor concentrations (0, 1, and 10 μM), with OD600 as the y-axis; (B) Growth curves of ΔFar1 ΔSst2 and wild-type BY4741 at different α-factor concentrations (0, 1, and 10 μM), with OD600 as the y-axis.
Explanation	To verify whether FAR1 and SST2 knockout relieves α-factor-induced growth inhibition, growth curves were measured at different α-factor concentrations, with wild-type BY4741 as the control. Wild-type BY4741 showed obvious growth inhibition at 10 μM α-factor, whereas ΔFAR1 maintained a relatively high growth rate under the same condition (Figure 1A), indicating that FAR1 knockout effectively relieved α-factor-induced G1 arrest. After further knocking out SST2 in the ΔFAR1 background, ΔFAR1 ΔSST2 showed no obvious growth inhibition at different α-factor concentrations compared with BY4741 (Figure 1B), suggesting that double knockout of FAR1 and SST2 further enhanced the strain's tolerance to α-factor.
Next plan	Fluorescence intensity curves will next be measured to evaluate the response of the ΔFAR1 ΔSST2 strains to α-factor induction, providing a basis for assessing the sensitivity and dynamic range of the biosensor.

No 14 Date：September 22–28, 2026 Experimenters: Xiaozhen Su Recorder: Xiaozhen Su

Experiment	Fluorescence Intensity Measurement of ΔFAR1 ΔSST2 Strains under α-factor Induction
Note	Protocol 16:Detection of Yeast Fluorescence Signal After Treatment

Result	 **Figure 1. Effect of Far1 and Sst2 knockout on α-factor-induced fluorescence signal.** (A) Fluorescence intensity curves of ΔFar1 and the yE-GFP control strain at different α-factor concentrations (0, 1, and 10 μM); (B) Fluorescence intensity curves of ΔFar1 ΔSst2 and the yE-GFP control strain at different α-factor concentrations (0, 1, and 10 μM).
Explanation	To verify the effect of Far1 and Sst2 knockout on downstream signaling, fluorescence intensity curves were measured at different α-factor concentrations. ΔFar1 showed a slight increase in fluorescence with increasing α-factor concentration, but the overall enhancement was limited (Figure 1A). In contrast, ΔFar1 ΔSst2 exhibited a more pronounced signal enhancement (Figure 1B): at 1 μM α-factor, its fluorescence was already much higher than that of the yE-GFP control, indicating that the knockout lowered the response threshold and enabled strong signaling at low inducer concentrations; at 10 μM α-factor, the fluorescence of the double-knockout strain continued to rise until 22 h without reaching a peak, whereas the control peaked at around 18 h and then declined, suggesting enhanced response amplitude and impaired negative feedback regulation. These results indicate that double knockout of Far1 and Sst2 enhances the yeast fluorescence response to α-factor, characterized by higher sensitivity at low concentrations and a more sustained response at high concentrations, supporting its potential as a low-concentration signal-amplifying biosensor element.
Next plan	Building on the ΔFar1 ΔSst2 strains obtained in this work, the next step will be to integrate them with the PAGER receptor module to validate their ability to detect influenza A virus HA antigen, and to further optimize the sensitivity and specificity of the biosensor.